

Notes: Abaluck & Gruber (RES, 2023)
When Less is More: Improving Choices in Health Insurance Markets

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1 Big picture

- **Question.** Does reducing health-insurance choice set size improve consumer choices, or does it make consumers worse off by removing valuable options?
- **Setting.** Oregon school districts choose subsets of state-negotiated health plans to offer employees.
- **Main descriptive fact.** The average employee could save roughly \$500–\$600 by switching to a lower-cost plan.
- **Core mechanism.** Consumers display inertia and approximate inertia: switchers move to nearby spreadsheet options rather than globally reoptimizing.
- **Policy result.** Smaller curated choice sets lower chosen costs; the 2012 reform that removed the four-plan cap increased costs.
- **Not choice overload.** Larger menus do not mainly make people worse choosers; they contain worse options that are not offset by reoptimization.

2 Incremental contribution of the paper

- Provides field evidence on choice-set size in a real health-insurance market.
- Studies curated choice restriction rather than random plan deletion.
- Introduces and estimates approximate inertia.
- Separates choice overload from menu composition.
- Compares choice-set restriction to equivalent information interventions.

3 Data and measurement

The paper uses OEBB enrollment and claims data from 2008–2012, district surveys, union contracts, district contributions, plan characteristics, and claims-based measures of expected out-of-pocket spending. The final analysis sample contains 92,133 beneficiaries. Gross premium is the plan price listed in plan documents. Net premium is $\max\{0, \text{Gross Premium} - \text{Employer Contribution}\}$. Residual premium is net premium minus gross premium. Foregone savings equals the cost of the chosen plan minus the lowest-cost plan in the choice set.

Interpretation: The key measurement innovation is to compute personalized plan costs under each alternative, then compare actual choices to lower-cost feasible choices.

4 Models and empirical specifications

4.1 Positive utility model

$$u_{ijt} = \beta_{0i}GP_{ijt} + \beta_{1i}RP_{ijt} + \beta_{2i}E(OOP)_{ijt} + \beta_{3i}Var_{ijt} + d_{jt} + \xi_{ij} + \theta_{ij} + \varepsilon_{ij}.$$

Here ξ_{ij} is inertia and θ_{ij} is approximate inertia.

Interpretation: The specification separates premium sensitivity, personalized cost sensitivity, ordinary inertia, and local spreadsheet switching.

4.2 Reduced-form choice-set regression

$$u_{itr}^N = \xi_J + x_{it}\gamma + \xi_{d,r} + \xi_{t,e(i)} + \varepsilon_{itr}.$$

The dependent variable is either foregone savings or total chosen costs. The coefficients ξ_J are dummies for number of plans, relative to two plans.

Interpretation: This estimates whether larger choice sets are associated with higher realized costs after controlling for district-tier effects, health-risk deciles, subsidies, and sample experience.

5 Identification and empirical design

Identification comes from variation in plan menus across districts and over time, especially the 2012 reform that removed the four-plan cap. Because premiums are centrally negotiated, district-level choice-set variation does not mechanically set plan prices. District-tier fixed effects compare changes within comparable local choice environments. Year-by-expenditure-decile fixed effects control for health-risk trends. The paper then tests whether the mechanism is choice overload by estimating choice models separately by number of plans and applying those choice functions to the same menus.

Interpretation: The central contrast is between demand-side deterioration and supply-side menu composition. The paper finds that menu composition is the primary driver.

6 Tables and figures

6.1 Table 1: Plans available by year; Table 2: Choice set by year

Read: Table 1 documents the plan universe over time. Table 2 shows that before 2012 most MODA beneficiaries faced 2–4 plans, while in 2012 many faced 5–7 plans.

Economic message: These tables establish the institutional menu variation behind the paper.

TABLE 1
Plans available by year

Plan	2008	2009	2010	2011	2012
Kaiser Medical Plan 1	Y	Y	Y	Y	Y
Kaiser Medical Plan 2	Y	Y	—	—	—
Kaiser Medical Plan 1A	—	Y	Y	Y	Y
ODS Medical Plan 3	Y	Y	Y	Y	Y
ODS Medical Plan 4	Y	Y	Y	Y	Y
ODS Medical Plan 5	Y	Y	Y	Y	Y
ODS Medical Plan 6	Y	Y	Y	Y	Y
ODS Medical Plan 7	Y	Y	Y	Y	Y
ODS Medical Plan 8	Y	Y	Y	Y	Y
ODS Medical Plan 9	Y	Y	Y	Y	Y
Providence Medical Plan 1	Y	Y	—	—	—
Providence Medical Plan 2	Y	Y	Y	Y	—
Providence Medical Plan 1A	—	Y	—	—	—
Providence Medical Plan 2A	—	—	Y	Y	—

Notes: 1. “Y” indicates that a plan was offered in at least one district and “—” indicates that a plan was not offered in a district in a given year. 2. Between 2012 and 2013 enrolment, ODS changed their name to MODA Health. Plans offer prior to 2013 and in 2013 are listed alphabetically, plans listed in the same row before and after the name change are not necessarily equivalent. In our analysis of plan choices in 2013, we consider whether plans introduced in 2013 we simply renamed or offered qualitatively different benefits.

TABLE 2
Choice set by year

Choice set size	2009	2010	2011	2012	All years
2	3,620	2,395	2,768	498	9,282
3	11,724	12,464	9,938	4,394	38,520
4	7,441	8,194	10,844	6,840	33,319
5	0	0	0	1,750	1,750
6	0	0	0	3,712	3,712
7	0	0	0	5,550	5,550

Notes: Table shows the total number of policy holders (including both individuals and families) enrolled in choice sets with the listed number of options in each year among beneficiaries who choose MODA plans. Prior to 2011, choice sets were limited to four plans and benefit managers chose up to four to offer to their employees. From 2012 on, benefit managers could choose between 1 and 10 plans to offer to their employees (only one small district offered a single plan, see row 7 from the analysis), including up to seven MODA plans.

(a) Table 1: Plans available by year. Original table directly cropped from the paper.

(b) Table 2: Choice set by year. Shows sample counts by menu size.

6.2 Figure 1: Mean foregone savings, MODA; Figure 2: Plan number changes for switchers

Read: Figure 1 shows substantial foregone savings under three expectations models. Figure 2 shows that actual switchers cluster around nearby plan numbers rather than cost-minimizing alternatives.

Economic message: The figures establish both the size of choice mistakes and the approximate-inertia mechanism.

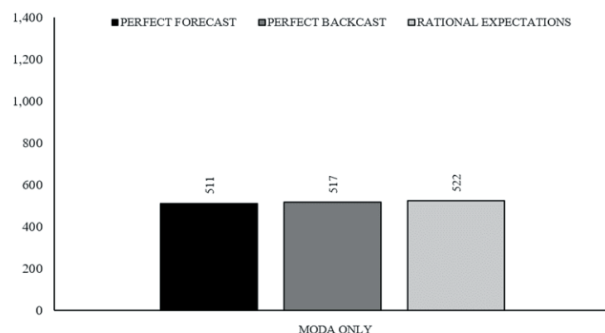


FIGURE 1

Mean foregone savings, MODA

Figure shows foregone savings among MODA plans using a variety of metrics to compute out of pocket costs. This is computed as the difference in costs between the chosen MODA plan and the lowest cost available MODA plan for beneficiaries who chose MODA. The perfect forecast, perfect backcast, and rational expectations models of out of pocket costs are described in the text.

(a) Figure 1: Mean foregone savings, MODA.

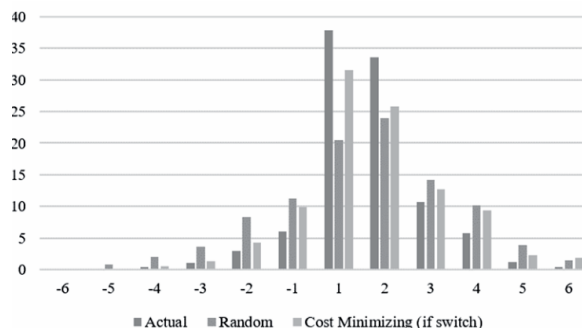


FIGURE 2

Plan number changes for switchers

Among beneficiaries who chose a MODA plan this year and the previous year but switched plans. "Actual" shows the distribution of the difference of the "number" of the plan chosen this year relative to the previous year. For example, a beneficiary who switched from MED3 to OMED4 would have a difference of 1. "Random" shows what this distribution would be if we randomly assigned beneficiaries alternative plans in their choice sets. "Cost Minimizing" shows what this difference would be if we assigned everyone to the cost minimizing plan (conditional on switching).

(b) Figure 2: Plan number changes for switchers.

6.3 Table 3: Logit models of plan choice (MODA only)

Read: Consumers respond more to gross premiums than residual premiums, underweight out-of-pocket costs, and display large inertia and approximate inertia.

Economic message: Table 3 is the core structural evidence of choice inconsistency.

	Perfect backcast	Perfect forecast	Rational expectations ¹
Gross premium (hundreds)	-0.073*** (0.006)	-0.073*** (0.006)	-0.073*** (0.006)
Residual premium (hundreds)	-0.026*** (0.004)	-0.026*** (0.004)	-0.026*** (0.004)
Mean OOP costs (hundreds)	-0.028*** (0.003)	-0.035*** (0.004)	-0.035*** (0.007)
Variance OOP costs (times 10 ⁶)	-	-	-0.006 (0.034)
Inertia	3700*** (82)	3702*** (82)	3706*** (82)
Approximate inertia	1957*** (64)	1958*** (65)	1962*** (65)
Deductible, in network (hundreds)	-0.048*** (0.001)	-0.046*** (0.001)	-0.046*** (0.001)
Max OOP, in network (hundreds)	-0.035*** (0.0005)	-0.034*** (0.0005)	-0.037*** (0.0005)
PCP copay, in network (hundreds)	-0.733*** (0.177)	-0.475*** (0.178)	-0.501*** (0.1)
Foregone welfare (Mean (SD))	517 (968)	511 (966)	522 (968)
Foregone welfare (no variance)	-	-	522 (970)
Percent selecting cost minimizing plan	47.6	47.6	47.6

Notes: 1. Rational expectations using regression predicted approach with 2,000 draws. 2. We drop beneficiaries with only 1 plan in their choice set. However, we do not drop beneficiaries with 1 MODA and 1 or more non-MODA plan, and thus a choice set of 1 when restricting to MODA plans. In Figure 1, we do drop such individuals to avoid having observation with mechanically 0 foregone savings. 3. *** Significant at the 1% level. ** Significant at the 5% level. * Significant at the 10% level.

Table 3: Logit models of plan choice (MODA only).

6.4 Figure 3: Foregone savings by choice set size; Table 4: Total costs vs. number of plans

Read: Foregone savings rise from roughly \$350 in two-plan menus to more than \$1,050 in seven-plan menus. Controlled regressions show seven-plan menus have \$742 more foregone savings and \$450 higher

total costs than two-plan menus.

Economic message: Smaller curated choice sets improve realized plan choices.

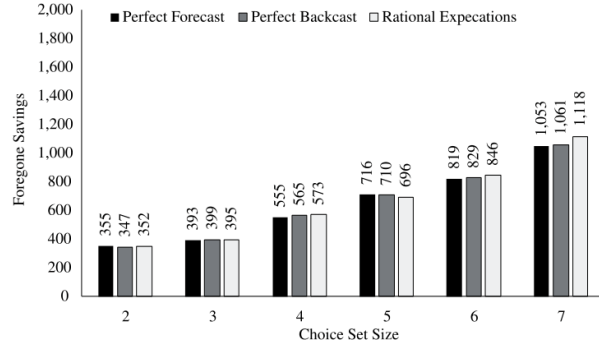


FIGURE 3
Foregone savings by choice set size (MODA only)

es: Figure shows the distribution of foregone savings by choice set size, counting only MODA plans.

(a) Figure 3: Foregone savings by choice set size.

TABLE 4
Total costs vs. number of plans

Sample	MODA only	
Metric	Foregone savings	Total costs
2	0	0
3	264	105
4	408	128
5	437	302
6	567	224
7	742	450
N	92,133	92,133

Notes: 1. Table shows results from a regression of foregone savings or total costs on dummies for the number of plans in the choice set controlling for choice set $\times$ tier $\times$ rate structure fixed effects, year $\times$ decile of expenditure fixed effects, and for the subsidy amount and the number of years in which the beneficiary appears in the data. 2. MODA only shows results from the regression considering only beneficiaries who chose MODA plans, the number of plans offered by MODA, and savings relative to the best MODA plan in the foregone savings column.

(b) Table 4: Total costs vs. number of plans.

6.5 Table 5: Simulated foregone welfare and total costs; Table 6: Logit model by number of plans

Read: These tables test choice overload. The choice function does not systematically deteriorate with menu size, and behavioral frictions persist across menu sizes.

Economic message: Larger menus are worse mainly because they contain worse plans, not because individuals become worse decision makers.

TABLE 5
Simulated foregone welfare and total costs

Sample	MODA only	
Metric	Foregone welfare	Total costs
Simulated—2 plans	550	2,785
Simulated—3 plans	535	2,770
Simulated—4 plans	502	2,736
Simulated—5 plans	479	2,713
Simulated—6 plans	506	2,740
Simulated—7 plans	555	2,790

Notes: The first row of Table 6 shows mean foregone savings and total costs in the “All Plans” and “MODA Only” samples. The remaining rows show results from a simulation constructed as follows. First, we estimate equation (1) allowing β , d_p , ξ , and θ to all vary flexibly with the number of available plans. This gives us a choice function that tells us how people choose from a given choice set. Each row of Table 6 uses the choice function associated with the listed number of plans to simulate choices from all choice sets.

(a) Table 5: Simulated foregone welfare and total costs.

TABLE 6
Logit model of plan choice (MODA only) by number of plans

	Number of plans					
	2	3	4	5	6	7
Gross premiums (hundreds)	−0.065*** (0.010)	−0.070*** (0.006)	−0.078*** (0.007)	−0.089*** (0.010)	−0.080*** (0.007)	−0.079*** (0.007)
Residual premium (hundreds)	−0.028*** (0.011)	−0.025*** (0.005)	−0.027*** (0.006)	−0.017 (0.012)	−0.030*** (0.006)	−0.029*** (0.006)
Mean OOP costs (hundreds)	−0.082*** (0.030)	−0.044*** (0.016)	−0.032*** (0.006)	−0.007 (0.028)	−0.018 (0.014)	−0.024* (0.013)
Inertia	3,973*** (219)	3,816*** (112)	3,432*** (135)	3,192*** (285)	−3,805*** (120)	−4,397*** (126)
Approximate inertia	1,782*** (353)	2,061*** (92)	1,865*** (87)	1,546*** (255)	1,776*** (154)	1,911*** (175)
Deductible, in network (hundreds)	−0.024*** (0.006)	−0.046*** (0.001)	−0.042*** (0.001)	−0.074*** (0.005)	−0.067*** (0.004)	−0.056*** (0.003)

Notes: 1. This table reports the logit coefficients which are used to generate the simulation results in Table 10. To obtain these coefficients, we estimate equation (1) separately by choice set size. This table is thus identical to column 3 of Table 3, with results broken out by choice set size. 2. *** Significant at the 1% level. ** Significant at the 5% level. * Significant at the 10% level.

(b) Table 6: Logit model by number of plans.

6.6 Figure 4: Probability offered vs. average costs; Tables 7–9

Read: Districts are less likely to offer high-cost plans. Larger menus raise premiums more than they improve out-of-pocket protection. Restricted and unrestricted districts look broadly similar.

Economic message: District benefit managers appear to curate menus by excluding high-cost plans.

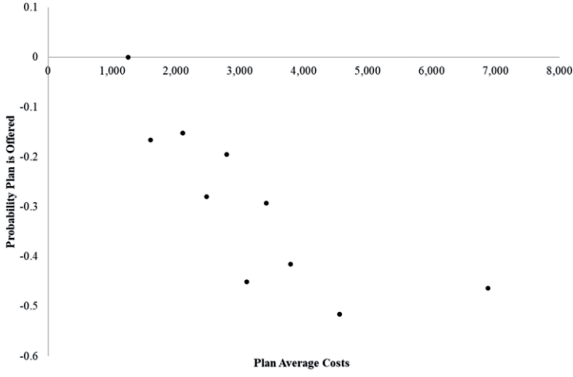


FIGURE 4
Probability offered vs. average costs (MODA only)

Notes: Figure shows plan offer probabilities by quantile of average costs. To construct this graph, we create an observation for each (year, district, plan, tier) and regress whether the plan is offered on average costs of that plan in that (year, plan, tier) as well as (year, tier) fixed effects. In other words, the regression is asking whether plans which are low cost relative to other plans available for that (year, tier) tend to be offered in more districts. The regression is normalized relative to the offer probability for (year, plan, tiers) in the first decile of average costs.

(a) Figure 4: Probability offered vs. average costs.

	Net premiums	Gross premiums	OOP
2	0	0	0
3	89	155	18
4	138	145	-14
5	187	21	107
6	229	243	-11
7	425	546	23

Notes: Table 7 parallels Table 4 but replaces total costs on the left-hand side of the number of plans regression with different components of total costs. These components are regressed on dummies for the number of plans in the choice set controlling for choice set $\times$ tier $\times$ rate structure fixed effects, year $\times$ decile of expenditure fixed effects, and for the subsidy amount and the number of years in which the beneficiary appears in the data.

(b) Table 7: What varies by choice set size?

	Restricted	Not restricted
Expenditures	11.294	10.582
Sex (% male)	28.5%	26.4%
Individual	0.283	0.250
Individual and child	0.229	0.233
Individual and spouse	0.120	0.120
Family	0.368	0.397

Notes: Comparison of average expenditures and beneficiary demographics for districts that did not offer (restricted) or did offer (not restricted) every available plan in 2012 and 2013 (pooled average), when benefit managers have the option of offering all available plans.

(a) Table 8: Expenditures and demographics by district restrictions.

Quantile	Enrolment size	Medical expenditure	Subsidy amount	% Families
2	-0.158 (0.329)	0.150 (0.256)	-0.033 (0.107)	0.011 (0.012)
3	-0.554 (0.418)	0.402 (0.279)	-0.120 (0.121)	0.004 (0.013)
4	-0.016 (1.057)	0.268 (0.268)	-0.119 (0.123)	0.008 (0.017)

Notes: Table reports regression coefficients from a single regression of the number of plans in each district on quantiles of number of beneficiaries enrolled, average expenditures in the district, subsidy amount and the percent of beneficiaries which are families as well as year, tier, and district fixed effects. Quantile 1 is the omitted category in each case.

(b) Table 9: Correlates of number of plans.

6.7 Table 10: Simulations: changing choice set size and providing information

Read: Restricting beneficiaries to best plans or removing worst plans often saves more than equivalent information interventions under the status quo.

Economic message: Choice-set restrictions survive inertia because removed plans cannot be selected; information must overcome inertia and approximate inertia.

	Year: best plan	District: best plan	Individual best plan	Year: remove two worst	District: remove two worst	Individual: remove two worst
Choice set intervention						
Status quo impact	215	332	537	75	245	377
Only approximate inertia	—	—	—	57	242	387
No inertia	—	—	—	19	227	378
Equivalent information intervention						
Status quo impact	164	192	204	95	175	197
Only approximate inertia	200	255	303	96	221	268
No inertia	211	329	510	71	248	375

Notes: The top panel shows simulations of the impact of changing choice sets on total costs in the chosen plan. The first row shows the estimated impact given the current degree of inertia, the second row the estimated impact given only approximate inertia, and the third row the estimated impact removing all inertia. The first column shows the impact of keeping only the best plan based on average costs among all beneficiaries in that year, the second column keeping only the best plan among all beneficiaries in that district, and the third keeping only the best plan for that individual (the degree of inertia is irrelevant with only one plan). The fourth column shows the impact of removing the two worst plans by average costs in that year, the fifth column the impact of removing the two worst plans by average cost in that district, and the sixth column the impact of removing the two worst plans for that individual. The bottom panel shows the impact of an information intervention (simulated as described in the Section 5.5. In this case, no plans are removed, but the information intervention is simulated so that, with no inertia, it is as effect as the choice set intervention in the status quo (this is done using a grid search for the scale parameter, and the sixth and first row do not match exactly due to the coarseness of this grid).

Table 10: Simulations of choice-set restrictions and information interventions.

7 Short summary

- Consumers leave substantial money on the table in health-insurance plan choice.
- The structural model reveals premium overweighting, out-of-pocket underweighting, inertia, and approximate inertia.
- Smaller curated menus produce lower costs.
- The simple choice-overload story is rejected; menu composition is the main mechanism.
- Choice-set restrictions can outperform information interventions because they are less muted by inertia.

8 Conclusion

8.1 Core conclusion

Less choice can be more when administrators can curate low-cost menus and consumers do not fully reoptimize.

8.2 Economic intuition

More plans increase heterogeneity benefits, but also expand opportunities for mistakes. Inertia and approximate inertia prevent consumers from avoiding bad added options.

8.3 Policy implication

Health-insurance exchanges and employers should consider curated choice architecture, not only information provision or maximal plan variety.

8.4 Limitations

The analysis is partial equilibrium and mainly within a single insurer's plans; full welfare would require pricing, competition, risk selection, and provider-network effects.

8.5 Takeaway

The paper shows that smart menu design can be a powerful substitute for consumer sophistication in complex insurance markets.